Ke Shi

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Education

University of Electronic Science and Technology of China — **Chengdu, China**
*B.S. in Computer Science (**Honors**); GPA: 3.91/4.00* — *Sep 2019 – Jun 2023*

University of Science and Technology of China — **Hefei, China**
Ph.D. student in Computer Science; Advisor: Shuai Shao; GPA: 4.11/4.30 — *Sep 2023 – Jun 2028 (Expected)*

Experience

ByteDance

Research Directions

Algorithms, combinatorial optimization, approximate counting, probabilistic combinatorics, and connections to phase transitions in statistical physics.

Publications

Zero-Freeness of the Hard-Core Model with Bounded Connective Constant
Yuan Chen, Shuai Shao, Ke Shi — *ISAAC 2026*

Strong Spatial Mixing for Continuous Spin Systems with Multi-Interactions
Shuai Shao, Ke Shi, Bohan Wang — *ISAAC 2026*

Zero-Freeness is All You Need: A Weitz-Type FPTAS for the Entire Lee–Yang Zero-Free Region
Shuai Shao, Ke Shi — *ITCS 2026*

An Optimal Algorithm for the Stacker Crane Problem on Fixed Topologies
Yike Chen, Ke Shi, Chao Xu — *ISAAC 2025*

Almost optimum ℓ-covering of $\mathbb{Z}_n$
Ke Shi, Chao Xu — *COCOON 2024*

A Polynomial Time Algorithm for Finding a Minimum 4-Partition of a Submodular Function
Tsuyoshi Hirayama, Yuhao Liu, Kazuhisa Makino, Ke Shi, Chao Xu — *SODA 2023 & Mathematical Programming 2023*

Open-Source Contributions

Google Summer of Code 2026 — **GeomScale/volesti**
Contributor, exact HMC for volume-based counting of linear extensions — *May 2026 – Aug 2026*

Open Source Promotion Plan 2025 — **Mogan STEM**
Mentor, Emoji Support Project — *Jul 2025 – Sep 2025*

Google Summer of Code 2024 — **GeomScale/dingo**
Contributor, modernized Dingo's linear-programming solver interface — *Jun 2024 – Aug 2024*

Open Source Promotion Plan 2024 — **Mogan STEM**
Contributor, OpenType MATH support — *Jul 2024 – Sep 2024*

Manuscripts

Strong spatial mixing for the matching polynomial: A simple proof via Heilmann–Lieb Theorem
Shuai Shao, Ke Shi — *2025, Submitted*

An LP Algorithm for Counting Eulerian Orientations Through the Lens of Quasi-polymorphism
Jincheng Guan, Shuai Shao, Ke Shi — *STOC '26 "Can AI Do Theory?" workshop; 2026, submitted*

Coupling Independence implies Zero-Freeness
Shuai Shao, Ke Shi

Selected Honors

National College Student Mathematics Competition — **Champion (Finals), 2023**
12th & 13th competitions: First Prize (Finals), 2021 & 2023